

CAREEROS

CareerOS

The evidence memory and action layer
between education and employment

ROUND 2 PRODUCT DOCUMENTATION

Talentbank AI Hackathon 2026 · July 2026

Remember the signal. Prove it. Act earlier.

Document Purpose and Reading Guide

This document replaces the Phase 1 product description with a Round 2 view grounded in the current CareerOS repository, implemented routes, local state flows, partial backend, and the deeper product-value analysis completed for the submission.

Truth boundary CareerOS is currently a polished multi-workspace prototype. It demonstrates product logic and rich interactions using local or mock data, deterministic recommendations, optional model calls, and a partial FastAPI backend. It does not yet prove adoption, customer demand, AI accuracy, production integrations, autonomous agents, or measured impact.

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1. Executive Summary

Career potential is frequently lost between moments. A student completes a project but cannot later explain what it proves. An employer meets strong early talent but restarts sourcing when an urgent role appears. A university identifies a capability gap after the current cohort has already lost time.

CareerOS is designed to preserve those signals as evidence and turn them into earlier decisions and actions. It connects three role-specific workspaces—Student, Employer, and University—through a common operating logic: signal → evidence → decision → action → outcome → learning.

Positioning CareerOS is the evidence memory and action layer between education and employment. It is more than a job portal, chatbot, dashboard, or isolated matching tool because its central value is continuity across decisions.

What changed in Round 2

- The student experience expanded beyond profiles and listings into Career Memory, Career Intelligence, skill development, opportunities, applications, interview practice, communities, mentorships, and an AI Companion.
- The employer experience expanded into an AI-first home, explainable talent discovery, candidate review, warm campus pipeline, engagement creation, hiring action queues, marketplace workflows, and analytics.
- The university experience expanded into Student Readiness, Curriculum–Market Alignment, Alumni Signals, Collaboration Marketplace, Accreditation Hub, and an AI Office with human decision controls.
- Cross-page stores now demonstrate selected evidence handoffs, shortlist and pipeline state, curriculum evidence packs, interventions, and accreditation readiness changes.
- The documentation now separates demonstrated, simulated/partial, and future-facing capabilities.

2. Product Thesis and Problem

2.1 One structural failure, three stakeholder consequences

Stakeholder	Failure today	Consequence
Student	Experience, feedback, and proof are fragmented.	Uncertainty, mass applications, random learning, and loss of confidence.
Employer	Early-talent interactions and decision context decay across events, spreadsheets, ATSs, and inboxes.	Repeated sourcing, weak validation, and lost warm candidates.
University	Readiness, market, alumni, partnership, and accreditation evidence arrives late or remains siloed.	Intervention after the current cohort can benefit and repeated evidence assembly.

2.2 Existing tools optimise transactions

Job boards optimise vacancies, ATS products optimise applicant processing, LMS products optimise course delivery, and reporting systems describe outcomes. These tools may remain important. CareerOS is designed to preserve the context between them so the next decision does not begin from zero.

2.3 Founder-level thesis

Core thesis CareerOS protects career potential from being lost between moments. Career Memory prevents evidence loss. Career Intelligence prevents option loss. Employer reactivation prevents relationship loss. University interventions prevent time loss. Accreditation reuse prevents institutional knowledge loss.

3. Users, Buyer, and Pilot Wedge

3.1 Primary users

Workspace	Representative user	Primary job
Student	Chris Lee, Year 3 Data Science student at Taylor's University	Preserve evidence, explore paths, build gaps, prepare, apply, and recover.
Employer	Recruiter, early-talent lead, campus-relations or hiring manager	Discover evidence-backed talent, validate fit, preserve relationships, and act on hiring priorities.
University	Faculty leader, programme director, career-services or employability owner	Detect readiness and market gaps, assign interventions, coordinate partners, and reuse evidence.

3.2 Recommended first buyer and wedge

The strongest initial buyer hypothesis is a faculty-level employability owner, sponsored by university leadership. The first product should not attempt to launch the entire three-sided network at once.

Recommended pilot One computing or data faculty, one cohort, two priority capability gaps, one career-services owner, and one or two employer-backed interventions over 8–12 weeks.

3.3 Why this wedge

- The university owns an urgent operational problem and can coordinate a bounded cohort.
- The workflow creates direct value before cross-organisation network effects exist.
- The intervention naturally activates student evidence and employer participation.
- Manual, consented data import is sufficient for an initial validation; full SIS, LMS, ATS, or HRIS integration is not required.

4. CareerOS Operating Model

4.1 Evidence-to-action sequence

Stage	CareerOS meaning	Examples
Signal	A moment that may matter later.	Project, challenge, application, hiring decision, readiness gap, alumni outcome.
Evidence	The signal with source, context, recency, and supporting material.	Career Memory entry, challenge observation, curriculum evidence, intervention record.
Decision	A prioritised judgement that exposes uncertainty.	Next skill, validation question, shortlist, intervention, programme priority.
Action	A specific follow-through state.	Practise, apply, re-engage, assign, collaborate, approve.
Outcome	The observable result of the action.	Completion, progression, candidate movement, intervention result.
Learning	Context preserved for the next decision.	Updated guidance, warmer pipeline, programme evidence, accreditation reuse.

4.2 Three workspaces, one product logic

Each workspace addresses a different expression of evidence loss, but they are not intended to be independent dashboards. Student evidence can support employer validation; employer interactions can create new student evidence; institutional gaps can trigger interventions that create both candidate signals and reusable university records.

4.3 What CareerGraph means in Round 2

CareerGraph is the product concept for connecting skills, experiences, roles, opportunities, employer demand, interventions, and outcomes. In the current prototype, this logic is represented through route-level interactions, local stores, mock datasets, and selected API calls. It is not yet a production shared graph database or live cross-organisation intelligence network.

5. Student Workspace

The Round 2 student experience is centred on a Data Science student and a continuous evidence-to-action journey rather than a one-time job search.

5.1 Career Memory

Career Memory preserves projects, internships, leadership, events, achievements, reflections, skills, and supporting evidence while the context is still available. Students can add or improve entries and see how experiences contribute to an evolving career story.

Designed behaviour change Maintain reusable evidence continuously instead of reconstructing a CV

only when an application appears.

5.2 Career Intelligence

Career Intelligence interprets existing evidence against possible paths, including adjacent roles beyond the degree title. It helps a student identify transferable strengths, missing proof, and a prioritised next step.

5.3 Skill development, opportunities, and applications

- Skill Development links an identified gap to a roadmap, learning activity, project, or evidence-building task.
- Opportunities explain why an event or role may fit and support saving, preparation, and application flows.
- Applications preserve stage, deadlines, and the next action across an application pipeline.

5.4 AI Companion and interview practice

The AI Companion provides intent-based guidance and routes the student to relevant context. Interview Practice supports role- and company-specific preparation, including voice-session interaction patterns. Optional backend/model paths exist, but scripted fallbacks and simulated content remain important parts of the demo.

5.5 Community, mentorship, and lifecycle return

Communities, mentorships, notifications, and recommendations create reasons to return between applications. The community experience is comparatively rich; mentorship and a complete rejection-to-recovery loop remain less mature and should be described accordingly.

6. Employer Workspace

6.1 Employer Home

The employer home acts as an AI briefing and action surface. It summarises operational signals, candidates, pipeline status, opportunities, and next actions without changing the underlying employer workflow.

6.2 Explainable talent discovery

Talent Discovery starts from a hiring need and presents candidate context, supporting evidence, strengths, gaps, and targeted validation questions. The interface models evidence-aware selection rather than opaque ranking.

Truth boundary Candidate records, match percentages, confidence indicators, and evidence histories are demo data. Production provenance, fairness evaluation, and validated matching accuracy are not implemented.

6.3 Candidate review and decision history

Candidate pages preserve shortlisting, pass decisions, interview preparation questions, and rationale through local state. The intended behaviour is to make decisions explainable and challengeable, rather than to treat a score as proof.

6.4 Campus pipeline and warm reactivation

Campus Pipeline demonstrates how prior challenges, events, workshops, or hiring processes could become a reusable warm pool. When an urgent need appears, employers can revisit known context instead of restarting sourcing from zero.

6.5 Engagements, marketplace, and analytics

- Engagement creation supports structured challenges, workshops, and campus activities designed to reveal applied ability.
- The marketplace and applicant workflows model role creation, applicant stages, and candidate movement.
- Analytics and briefing views show how employer activity could create reusable operational learning; displayed outcomes remain illustrative.

7. University Workspace

7.1 AI-first university overview

The Round 2 overview frames the university as an employability operating system. It surfaces programme, cohort, market, partnership, accreditation, and intervention signals through action-oriented cards and natural-language routing.

7.2 Student Readiness

Student Readiness highlights hidden employability risk, differentiates academic performance from career readiness, and supports assignment of intervention owners, deadlines, and follow-up states.

7.3 Curriculum–Market Alignment

Curriculum–Market Alignment compares programme coverage with illustrative demand signals, provides evidence chains, and supports current-cohort interventions while formal curriculum changes continue. Evidence packs and roadmap actions demonstrate how analysis becomes an operational artefact.

7.4 Alumni Signals and Collaboration Marketplace

Alumni Signals route graduate outcomes and employer feedback back into programme discussions. Collaboration Marketplace connects identified capability gaps to employer, society, workshop, challenge, and outreach workflows.

7.5 Accreditation Hub

Accreditation Hub models continuous evidence readiness, named ownership, requirement tracking, and reuse across MQA, SETARA, QS, and internal review contexts. It demonstrates preparation and reuse—not automated accreditation submission.

7.6 University AI Office

The AI Office and Decision Room present specialised assistant roles that prepare cross-department actions. Human approval remains explicit. Some optional model calls are configured in the frontend, but the operating-room intelligence, metrics, and external effects are simulated.

8. Cross-Workspace Evidence Loop

8.1 Example: one capability gap

1. A university identifies a cloud or MLOps readiness gap affecting the current cohort.
2. The programme team selects a targeted employer-backed challenge or workshop while curriculum reform proceeds.
3. Students participate and demonstrate applied ability.
4. The activity can become Career Memory evidence with source and context.
5. An employer can use the evidence as a discovery and validation signal.
6. The university measures participation and follow-through, then reuses the record in programme and accreditation work.

Round 2 distinction The current repository demonstrates many components of this loop and selected state handoffs. The complete cross-organisation evidence network, permissions, and outcome propagation remain future-facing.

8.2 Why the loop matters

Stakeholder	Value that can accumulate
Student	A portable career story and clearer evidence gaps.
Employer	Relationship history, observed signals, and reusable candidate context.
University	Intervention history, outcome learning, and reusable institutional evidence.
Talentbank	A potential lifecycle relationship beyond listings and one-off applications.

9. Implementation Status

Capability	Current status	Boundary
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Capability	Current status	Boundary
Three role-based workspaces	Demonstrated	Protected routes and rich local interactions are implemented.
Career Memory and profile state	Demonstrated / partial	Local state and selected extraction paths; production verification is incomplete.
Career Intelligence and paths	Demonstrated / simulated	Interactive paths and recommendations use illustrative data.
Opportunities and applications	Demonstrated locally	No live job marketplace or production application exchange.
Interview practice and AI Companion	Demonstrated / optional AI	Backend or model paths may be used; fallbacks and scripted states remain.
Communities and mentorships	Mixed	Community is richer; mentorship lifecycle is thinner.
Employer discovery and pipeline	Demonstrated / simulated data	No validated matching, fairness proof, or production ATS integration.
University readiness and curriculum workflows	Demonstrated / simulated data	Signals, scores, and impact values are illustrative.
Accreditation and AI Office	Demonstrated interaction model	Formal integrations, autonomous action, and production governance are future.
Cross-workspace network	Partial concept proof	Selected stores and handoffs exist; live shared intelligence does not.

9.1 Claims removed or corrected from Round 1

- Removed unsupported market-size and prevalence statistics.
- Replaced “verified evidence” with evidence-oriented or verification-model language.
- Removed the claim that Google ADK multi-agent orchestration is fully implemented across the product.
- Removed production Cloud Run, Cloud SQL, OAuth, SIS/LMS/ATS, and labour-market integration claims.
- Replaced predictive accuracy and measured-impact language with designed mechanisms and pilot metrics.
- Reframed competitive uniqueness and business-model claims as hypotheses requiring validation.

10. Technical Architecture

10.1 Frontend

Layer	Current implementation
Framework	React 19 with Vite 5.
Routing	React Router with role-protected Student, Employer, and University routes.

Layer	Current implementation
Styling	Tailwind CSS plus application-specific styles.
State	Zustand stores and local component state for selected cross-page workflows.
Visualisation	React Force Graph and custom React components.
Icons	Lucide React.

10.2 Backend

The repository contains a partial FastAPI and SQLAlchemy backend with candidate, employer, and user routers, database models, migrations, and utilities. Candidate-facing service code references chat, interview-session, and add-experience paths. The frontend also includes local fallbacks when a backend endpoint or environment configuration is unavailable.

10.3 Intelligence paths

- Deterministic and mock datasets drive many recommendations, scores, summaries, and workflow states.
- Selected frontend services can call candidate chat, interview, and experience-extraction endpoints.
- The University AI Office contains an optional direct model configuration path.
- Backend agent code and sub-agent experiments exist, but they should be treated as partial technical exploration rather than a complete production orchestration layer.

10.4 Production requirements

A production release would require secure identity, role and consent enforcement, durable storage, audit logs, provenance, data minimisation, integration contracts, model evaluation, observability, deployment hardening, and institutional governance.

11. AI, Data, and Governance

11.1 AI product principle

AI should reduce navigation, synthesise context, and prepare decisions. It should not conceal uncertainty, substitute an opaque score for evidence, or execute consequential external actions without authority.

11.2 Human-governed interaction

AI may prepare	Humans must control
A next-action recommendation	Whether the student accepts or shares it.
Candidate fit rationale and validation questions	Shortlist, rejection, interview, and hiring decisions.
Intervention or curriculum options	Programme priorities, owners, budgets, and implementation.
Accreditation evidence drafts	Formal claims, approval, and submission.

AI may prepare	Humans must control
Cross-department action brief	Any external or consequential action.

11.3 Data and evidence requirements

- Student ownership, consent, correction, revocation, and visibility rules for each Career Memory item.
- Source, provenance, recency, confidence, and verification state for evidence.
- Clear separation between observed evidence, inferred recommendation, and user-entered claim.
- Role-based access, minimisation, retention, appeal, and auditability.
- Fairness and outcome validation before predictive or ranking claims.

12. Strategic Value for Talentbank

CareerOS could extend Talentbank’s relevance from individual career transactions into a longer evidence-and-action lifecycle.

Dimension	Potential strategic value	What must be validated
Student lifecycle	Engagement before, during, and after applications.	Retention triggers, consent, and sustained evidence maintenance.
Employer offering	Discovery, warm pipeline recovery, engagement design, and reusable talent context.	Buyer urgency, trust, supply, and workflow integration.
University offering	A faculty-level employability operating layer.	Budget ownership, implementation capacity, and measurable workflow value.
Network learning	Challenges and interventions create new evidence and relationships.	Legal rights, data-sharing boundaries, provenance, and governance.
Defensibility	Longitudinal context and operational history may become harder to replicate.	Real adoption, permissioned data, outcomes, and distribution.

Strategic boundary The repository does not prove Talentbank partnerships, proprietary data access, customer demand, or the right to combine student, employer, and university records. Strategic fit is a hypothesis for leadership validation.

13. Pilot and Validation Plan

13.1 Pilot sequence

7. Select one faculty, cohort, and employability owner.
8. Agree on two safe, actionable capability gaps.

9. Import a minimum controlled cohort dataset with explicit consent and manual verification.
10. Run one or two employer-backed interventions.
11. Capture student participation and evidence updates.
12. Review employer validation and warm-pipeline usefulness.
13. Produce one faculty action and evidence summary.
14. Conduct governance, fairness, and user-trust review before expansion.

13.2 Metrics

Area	Pilot metric
Activation	Eligible-student activation and Career Memory evidence completion.
Action	Intervention uptake, completion, and time from gap detection to action.
Employer	Partner participation, qualified challenge performers, and warm-candidate follow-up.
University	Staff time to produce an action/evidence summary and follow-through completion.
Trust	Consent comprehension, evidence correction requests, user confidence, and fairness concerns.
Retention	Return behaviour after deadlines, practice, rejection, intervention, or new evidence.

13.3 Success criterion

Prove the workflow before the network The pilot should demonstrate that preserved evidence leads to a clearer next decision and completed action. It should not attempt to prove national-scale intelligence, predictive accuracy, or a complete marketplace.

14. Product Roadmap

Stage	Product focus	Exit criteria
Now — Round 2 prototype	Stabilise three workspaces, reliable demo paths, truth labels, and selected handoffs.	Consistent build, reliable flows, no unsupported claims, known fallback behaviour.
Pilot readiness	Consent, role access, provenance, auditability, controlled import, and pilot instrumentation.	Approved data model, governance review, pilot owner, baseline measures.
Bounded pilot	Faculty intervention loop with employer participation and student evidence.	Measured activation, follow-through, staff effort, trust, and workflow usefulness.
Post-pilot expansion	Deeper readiness, collaboration, employer pipeline, alumni, and accreditation workflows.	Evidence that each expansion solves a paid operational problem.

Stage	Product focus	Exit criteria
Future network	Permissioned integrations and longitudinal outcome learning.	Secure integration, validated models, clear governance, and demonstrated demand.

14.1 Priority product improvements

- Add persistent demo-data and simulated-action labels to scores, sends, exports, and agent outputs.
- Complete one challenge → Career Memory → employer profile → university outcome story.
- Add evidence provenance and recency to employer-visible context.
- Strengthen rejection-to-diagnosis and mentorship-to-Memory loops or keep them secondary.
- Make current-cohort intervention explicit on Curriculum–Market Alignment.
- Show accreditation reuse with source, period, owner, and validity.
- Clarify source, prepared action, required approval, and external effect inside the AI Office.

15. Setup and Demo Guidance

15.1 Frontend

- Install dependencies: `npm install`
- Start development: `npm.cmd run dev`
- Production build check: `npm.cmd run build`
- Open the local Vite URL and choose the intended workspace through the landing flow.

15.2 Backend

The backend is partial and environment-dependent. Configure database and model credentials locally before running FastAPI services. Demo planning should define whether the frontend uses a live endpoint or its deterministic fallback and avoid switching state during a judged walkthrough.

15.3 Recommended stable demo

15. Student: Career Memory → Career Intelligence → one preparation or opportunity action.
16. Employer: evidence-aware candidate detail → targeted validation → Campus Pipeline reactivation.
17. University: curriculum/readiness gap → current-cohort intervention → evidence pack / Accreditation Hub.
18. AI Office close: prepared action with explicit human approval and simulated-status language.

15.4 Avoid in the main demo

Avoid placeholder settings/help routes, legacy aliases, unsupported exports, unconfigured backend calls, generic metrics without provenance, and any interaction that implies a live external send or production integration.

Appendix A. Route and Capability Inventory

Workspace	Current primary routes
Student	/student/home; /student/profile; /student/intelligence; /student/opportunities; /student/applications; /student/communities; /student/mentorships; /student/skill-development; /student/ai-companion
Employer	/employer/home; /employer/talent; /employer/talent-discovery; /employer/posting; /employer/campus-pipeline; /employer/analytics; /employer/candidates; /employer/marketplace
University	/university/overview; /university/student-readiness; /university/curriculum-alignment; /university/alumni-signals; /university/collaboration; /university/accreditation
Legacy / placeholders	Several redirects and placeholder settings, reports, learning, and help routes remain in the repository and should not define the product narrative.

Backend surface observed

FastAPI includes user, candidate, and employer routers in backend/app/app.py. Additional router files and agent experiments are present, but not every file is included in the active application or connected to the Round 2 frontend.

Appendix B. Claim Language

Avoid	Use instead
Verified evidence	Evidence-oriented profile; the interface models verification and provenance.
AI predicts success	AI-supported recommendation; predictive accuracy is not validated.
Shared intelligence network	The architecture proposes a shared network; current workflows mostly use local or mock state.
Saves time / improves ROI	Designed to reduce repeated work; measure time and economic value in a pilot.
Autonomous AI employees	Specialised assistant interaction with human approval; autonomy is future-facing.
Automated accreditation	Accreditation readiness, evidence reuse, ownership, and draft preparation.
Talentbank advantage	Potential strategic fit pending validation of assets, rights, demand, and distribution.
Customers / partnerships / adoption	Do not claim unless documented evidence is provided.

Source basis

- CareerOS Deep Value Discovery, docs/pitch/CAREEROS_DEEP_VALUE_DISCOVERY.md.
- Round 1 submission, [Team GongBaiWan] CareerOS_Documentation_V1.pdf.
- Refined editable documentation source, CareerOS_Product_Documentation_Refined.docx.
- Current repository routes, components, stores, services, mock data, and backend files inspected in July 2026.